Matgeo Presentation - Problem 5.5.6

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October 11, 2025

Problem Statement

For $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$, then $14\mathbf{A}^{-1}$ is given by

Solution

Given

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} \tag{0.1}$$

Let $\mathbf{A}^{-1}$ be the inverse of $\mathbf{A}$. Then

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} \tag{0.2}$$

Augmented matrix of $\left(\mathbf{A} \;\middle|\; \mathbf{I}\right)$ is given by

$$\left(\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{array}\right) \tag{0.3}$$

Applying elementary row transformations:

$$\left(\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \to 3R_2 + R_1} \left(\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ 0 & 7 & 1 & 3 \end{array}\right) \tag{0.4}$$

Solution

$$\left(\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ 0 & 7 & 1 & 3 \end{array}\right) \xrightarrow{R_1 \to 7R_1 - R_2} \left(\begin{array}{cc|cc} 21 & 0 & 6 & -3 \\ 0 & 7 & 1 & 3 \end{array}\right) \tag{0.5}$$

$$\xrightarrow{R_1 \to \frac{1}{21}R_1, R_2 \to \frac{1}{7}R_2} \left(\begin{array}{cc|cc} 1 & 0 & \frac{2}{7} & -\frac{1}{7} \\ 0 & 1 & \frac{1}{7} & \frac{3}{7} \end{array}\right) \tag{0.6}$$

Hence the inverse of matrix **A** is

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{2}{7} & -\frac{1}{7} \\ \frac{1}{7} & \frac{3}{7} \end{pmatrix} \tag{0.7}$$

Therefore, $14\mathbf{A}^{-1}$ is given by

$$14\mathbf{A}^{-1} = 14 \times \begin{pmatrix} \frac{2}{7} & -\frac{1}{7} \\ \frac{1}{7} & \frac{3}{7} \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ 2 & 6 \end{pmatrix} \tag{0.8}$$